STALKS OF FLAT MODULES ARE FLAT: \MATHCAL{F} FLAT \RIGHTARROW \MATHCAL{F}_P FLAT OVER \MATHCAL{O}_P

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one same-chapter section pointer and one same-chapter lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Pullback of modules preserves flatness. Stacks Project, Modules on Sites, Lemma lemma-pullback-flat. Supplies that p\^* of a flat \mathcal{O}-module is flat over \mathcal{O}_p.
- Points as morphisms of ringed topoi. Stacks Project, Modules on Sites, section on skyscraper sheaves. Supplies the morphism (p, id_{\mathcal{O}_p}) whose pullback functor is the stalk functor.

Source reference. Original section 05V6: sources/stacks-chow/tags.txt, source lines 7540–7540. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05VD: sources/stacks-chow/tags.txt, source lines 7547–7547. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let p be a point of $\mathcal{C}$. If $\mathcal{F}$ is a flat $\mathcal{O}$-module, then $\mathcal{F}_p$ is a flat $\mathcal{O}_p$-module.*

Proof. In Section 05V6 we have seen that we can think of p as a morphism of ringed topoi

$$(p, \mathrm{id}_{\mathcal{O}_p}) : (Sh(pt), \mathcal{O}_p) \longrightarrow (Sh(\mathcal{C}), \mathcal{O}).$$

such that the pullback functor $p^* : Mod(\mathcal{O}) \to Mod(\mathcal{O}_p)$ equals the stalk functor. Thus the lemma follows from Lemma 05VD. □

Date: stacks-project:sites-modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.